AMIT BAKALE

EMBEDDED SOFTWARE ENGINEER

Embedded C/C++ • Embedded Linux • RTOS • Avionics & Automotive Systems

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PROFESSIONAL SUMMARY

Results-driven Embedded Software Engineer with 3+ years of hands-on experience developing safety-critical software for Aerospace/Avionics (Teledyne Controls ECP) and Automotive IoT/Telematics. Specialized in Embedded C, modern C++ (C++11/14/17), Embedded Linux (Yocto), RTOS (FreeRTOS), and Multi-Agent AI Frameworks. Proven expertise delivering dual-bank (A/B) flash failover daemons, RabbitMQ/AMQP messaging, OpenSSL/PKI security (EAP-TLS, FreeRADIUS), and multi-bus protocols (ARINC 429/717, CAN, TCP/IP, MQTT). Adept at board bring-up, custom bootloader design, robust OTA updates, and containerized edge targets.

TECHNICAL SKILLS

Programming Languages: Embedded C, C++ (C++11/C++14/C++17), Python, Shell Scripting (Bash)

Embedded & Operating Systems: Embedded Linux (Yocto, Linux Kernel basics), RTOS (FreeRTOS), Device Drivers, Custom Bootloader, U-Boot, Dual-Bank (A/B) Flash, Board Bring-Up, POSIX Multithreading (Pthreads, Mutex, Semaphores), Memory Management

Avionics & Automotive Systems: ARINC 429, ARINC 717, CAN / CAN-FD, Electronic Flight Bag (EFB) Connectivity, Telematics & Diagnostics (MNVR), ADAS Concepts, DO-178C Traceability Principles, Fail-Safe Dual-Bank Recovery

Protocols & Networking: TCP/IP, UDP, POSIX Sockets, AMQP (RabbitMQ), MQTT, UART, SPI, I2C, RS-485, USB, Ethernet, VLAN Slicing, Network Routing, cURL

Security, PKI & Cryptography: PKI, OpenSSL, X.509 Multi-Trust Chains, CRL Validation, PKCS#12, CMS Verification/Signing, FreeRADIUS, EAP-TLS Mutual Authentication, Cryptographic Hashing (MD5, SHA-256)

DevOps, Tools & Hardware: Docker (Build/Save/Deploy), CMake, GCC/G++, GDB, Git/GitHub, Linux CLI/Systemd, Valgrind, STM32, ESP32, D15 Control Card, ST-Link, Logic Analyzers, Putty, MobaXterm

Agentic AI & Engineering Automation: Multi-Agent AI Frameworks (Requirements, Architecture Auditing, DO-178C Traceability, Automated Code Review, Static Analysis & Gap Assessment)

PROFESSIONAL EXPERIENCE

NATRINAI TECHNOLOGIES | Chennai, Tamil Nadu

Embedded Software Engineer | Feb 2026 – Present

Teledyne Controls ECP Configuration Manager (config_mgr) – Avionics Edge Platform

- **Avionics Configuration Daemon:** Developed and optimized the safety-critical config_mgr C++ daemon for the Teledyne Controls Electronic Computing Platform (ECP), acting as the central authority governing active (/work), standby (/stby), and default configurations on onboard flash storage.
- **Dual-Bank Failover & Flash Integrity:** Engineered dual-bank flash partition fallback mechanisms using cryptographic MD5 digest verification, ensuring automatic rollback to standby partitions and eliminating boot-looping/flash corruption risks during flight power interruptions.
- **Decoupled AMQP Messaging & Subsystem Slicing:** Implemented asynchronous RabbitMQ (AMQP 0-9-1) inter-process messaging across POSIX threads, slicing master configurations and dynamically routing payloads to avionics modules: ADIF (ARINC 429/717, discrete inputs), AIO, NIC (VLANs, routing), HAP, and Health Manager.
- **Air-to-Ground (A2G) Synchronization:** Built bidirectional cloud downlink/uplink pipelines with JSON Schema (v3.36) validation, ground ACK confirmation events, and dynamic 1-hour quiet-period debounce algorithms for gate/flight transitions.

Config Manager Multi-Agent AI Framework – Autonomous Systems Engineering

- **Autonomous Engineering Pipeline:** Architected an end-to-end multi-agent AI framework spanning 9 lifecycle phases (Requirements, Architecture, Planning, Coding, Automated Code Review, Unit Testing, Release, Integration, and Orchestration) for embedded avionics software.
- **DO-178C Traceability & Gap Analysis:** Developed agentic workflows mapping software requirements directly to verification test harnesses, generating traceability matrices and automating engineering gap analysis aligned with DO-178C assurance principles.
- **Automated Refactoring & Static Analysis:** Utilized AI agents to audit legacy codebases, eliminate technical debt (blocking sleep reconnect loops, singleton coupling), and formulate migration pathways to modern C++17/20 concurrency.

Network Authentication Function (NAF) – PKI & Edge Security

- **Multi-Trust PKI Architecture (ECP-1124):** Designed multi-trust CA directory architectures enabling mutual EAP-TLS authentication for Electronic Flight Bags (EFBs) across multiple commercial airline operators connecting to onboard edge access points.
- **Cryptographic Validation Engine:** Integrated OpenSSL for automated X.509 certificate validation, CRL (Certificate Revocation List) tracking, PKCS#12 bundle extraction, and Cryptographic Message Syntax (CMS) signature verification.
- **FreeRADIUS Integration & Diagnostics:** Integrated FreeRADIUS services on embedded Linux; diagnosed EAP-TLS handshake anomalies using radiusd internal diagnostics, OpenSSL inspection, and dynamic credential bundle management.
- **Containerized Edge Deployment:** Orchestrated Docker build-to-target deployment workflows (image tar compression, MD5 manifest integrity updates, remote SCP sync, and systemd daemon management) on Yocto embedded hardware.

PRIME EDGE INFO SOLUTIONS PVT. LTD. | Chennai, Tamil Nadu

Embedded Software Developer | Aug 2024 – Jul 2025

- **MNVR Real-Time Telematics Baseboard:** Developed Embedded C/C++ firmware on MC6630/MCP and STM32 microcontrollers for real-time fleet telematics and smart-city bus telemetry acquisition.
- **Custom Bootloader & Robust OTA:** Designed custom dual-stage bootloader and fail-safe OTA update mechanisms over network/UART, ensuring 100% firmware recovery during transmission dropouts.
- **Multi-Sensor & CAN Integration:** Interfaced vehicle Bus CAN, UART, GPS module, temperature sensors, GPIO, SOS emergency triggers, tamper detection, and ignition inputs for real-time edge processing.
- **IoT Cloud Connectivity:** Built lightweight socket communication and MQTT/TCP-IP telemetry streaming pipelines to send real-time vehicle telemetry to municipal cloud servers.
- **Smart LED Display System (Nashik Smart City):** Engineered embedded C/C++ firmware on D15 Control Card under RTOS, utilizing tasks, semaphores, and message queues to process dynamic ETA and route displays.
- **High-Throughput Network Fetching:** Implemented TCP/IP, UDP, and cURL communication protocols for seamless cloud schedule synchronization and automated diagnostics.
- **ESP32 Edge Gateway:** Developed ESP32 firmware for P10 LED destination boards supporting RS-485 industrial serial bus, non-volatile Flash storage, and Python-driven automated validation scripts.

EDUCATION & CERTIFICATIONS

B.E. Computer Science & Engineering

Visvesvaraya Technological University (VTU)
KLE College of Engineering & Technology, Chikodi | 2019 – 2023

Certified Embedded Professional (ECEP)

EMERTXE – Institutes for Advanced Embedded Systems
Jun 2023 – Jul 2024 | Embedded Linux, firmware, networking, hardware debugging

KEY ACCOMPLISHMENTS & CORE COMPETENCIES

- **Safety-Critical Avionics & Edge Platforms:** End-to-end design of Teledyne Controls ECP configuration daemons, dual-bank fail-safe partitioning, ARINC 429/717 data slicing, and DO-178C-aligned traceability.
- **Automotive & Commercial Telematics:** Board bring-up, custom bootloader design, CAN bus protocol integration, STM32/ESP32 firmware, and multi-sensor acquisition across fleet deployments.
- **Enterprise Edge Security & Cryptography:** Architected multi-trust EAP-TLS authentication for Electronic Flight Bags (EFBs), OpenSSL X.509/CRL validation, and FreeRADIUS directory management.
- **Modern AI-Driven Systems Engineering:** Pioneer in multi-agent autonomous engineering frameworks for automated code review, static code debt remediation, and C++ modernization.

LANGUAGES

English (Professional) | Kannada (Native) | Hindi (Fluent) | Marathi (Fluent)